

SmartRoute AI — Technical Paper

Customer Query Routing Classification — Capstone Problem 35

Abstract

Customer-support systems receive large volumes of natural-language requests that must be classified and routed to suitable service functions. SmartRoute AI uses TF-IDF text representation and Logistic Regression for multi-class intent classification, then applies a confidence-aware routing policy through a Flask web application.

Metric	Result
Accuracy	87.24%
Weighted Precision	88.02%
Weighted Recall	87.24%
Weighted F1-score	87.26%

1. Problem Statement

Classify incoming customer queries into predefined service categories so that requests can be routed to an appropriate service queue.

2. Dataset

BANKING77. The final local run used 10,003 training examples, 3,080 official test examples, and 77 intent classes.

3. Methodology

Input validation → TF-IDF feature extraction → Logistic Regression → intent prediction → confidence-aware routing → service queue → browser history and analytics.

4. EDA Findings

The final EDA run reported $10,003 \times 2$ training shape, $3,080 \times 2$ test shape, columns text/category, zero missing values, zero duplicate rows in train/test, zero exact train/test text overlap, and 77 classes. Mean training query length was approximately 11.95 words and 59.48 characters.

5. Evaluation

The final local official-test run produced the metrics below.

6. Prototype

The application provides customer query input, prediction, confidence, top predictions, routing decision, service queue, routing history, and analytics. History is stored in browser localStorage for demonstration.

7. Limitations

TF-IDF + Logistic Regression can struggle with semantically similar intents and unusual phrasing. Routing thresholds are application rules and should not be interpreted as calibrated probability guarantees.

8. Future Work

Probability calibration, human-in-the-loop feedback, multilingual support, drift monitoring, richer metadata, and transformer-based comparisons.

9. Conclusion

SmartRoute AI demonstrates an end-to-end, reproducible customer-query routing prototype using conventional NLP and machine learning.